

EXP 13**Case Study on Low-Code Limitations****AIM**

To study the limitations of low-code development platforms using Zoho Creator and analyze their impact on application development, customization, performance, integration, and scalability.

PROBLEM STATEMENT

A hotel wants to develop a booking management system using Zoho Creator. The application should manage customer details, room availability, reservations, payments, cancellations, notifications, and reports.

Zoho Creator enables rapid application development through visual tools, predefined components, workflows, and Deluge scripting. However, as business requirements become more complex, certain limitations of low-code development may affect flexibility and scalability.

SYSTEM REQUIREMENTS

- Customer registration and login
- Room availability and booking management
- Payment processing
- Booking confirmation and cancellation
- Email and SMS notifications
- Reports and dashboards
- Integration with external hotel management systems

PROCEDURE:

1. Log in to Zoho Creator and create a sample application, such as a Hotel Booking Management System.
2. Create forms for customer details, room information, bookings, and payments using the drag-and-drop interface.
3. Configure workflows and business rules using Deluge scripting.
4. Identify limitations in advanced customization, complex logic, third-party integrations, and UI design.
5. Test application performance with increasing numbers of records and users.
6. Examine integration requirements and platform-specific dependencies.
7. Analyze the limitations and identify suitable workarounds or alternative technologies.
8. Document the observations and conclude the suitability of low-code development for the application.

LIMITATIONS IDENTIFIED

1. Limited Customization

Low-code platforms provide predefined UI components and templates. Creating a completely unique interface or implementing highly customized user interactions may require workarounds or external development.

2. Complex Business Logic

Simple conditions and workflows are easy to implement. However, complex pricing rules, multiple booking dependencies, and advanced reservation algorithms may require extensive Deluge scripting or external services.

3. Platform Dependency (Vendor Lock-in)

Applications developed using Zoho Creator depend on its platform-specific tools, data models, and Deluge language. Migrating the entire application to another platform may require rewriting scripts and redesigning workflows.

4. Performance and Scalability

As the number of bookings, customers, and reports increases, inefficient queries, excessive workflow execution, or platform resource limits may affect performance. Large-scale applications require careful optimization and testing.

5. Integration Limitations

Integrating external payment gateways, hotel reservation systems, or third-party APIs may require API configuration, authentication, custom functions, and additional development.

6. Limited Control Over Infrastructure

The platform manages much of the underlying infrastructure. Developers have less direct control over server configuration, database optimization, deployment architecture, and low-level system settings than with a fully custom application.

7. Security and Compliance Considerations

Although Zoho Creator provides security features, developers must correctly configure permissions, record access, authentication, and data sharing. Specific legal or industry compliance requirements may need additional controls and verification.

8. Licensing and Cost

Advanced features, higher usage, additional users, or greater resource requirements may affect subscription costs. Organizations should evaluate pricing and plan limits before deploying large applications.

PROPOSED SOLUTION

Limitation	Proposed solution
Limited customization	Use custom pages, supported HTML/CSS, and integrations where appropriate
Complex business logic	Use optimized Deluge scripts or external services
Vendor lock-in	Document workflows and maintain exportable data backups
Performance issues	Optimize reports, queries, and workflows
Integration challenges	Use supported APIs, webhooks, and connectors
Infrastructure restrictions	Use external services when greater control is required
Security concerns	Apply least-privilege permissions and regularly audit access
Cost concerns	Review usage, licensing, and resource requirements

OBSERVATION

S. No.	Observation	Result
1	Forms created using drag-and-drop tools	Faster development
2	Basic workflows configured	Less manual coding
3	Complex logic implemented	Additional scripting may be required
4	External integrations examined	API configuration may be necessary
5	Performance requirements analyzed	Optimization becomes important as usage grows
6	Security permissions configured	Access can be controlled through appropriate settings
7	Platform dependency evaluated	Migration may require redevelopment
8	Licensing requirements reviewed	Cost planning is necessary

RESULT

The limitations of low-code development in Zoho Creator were studied through the Hotel Booking Management System case study. The analysis identified challenges related to customization, complex business logic, integration, performance, scalability, security, cost, and platform dependency. Suitable solutions were proposed to reduce their impact.

CONCLUSION

Thus, the case study on low-code limitations in Zoho Creator was completed successfully. Low-code platforms enable rapid application development, reduce manual coding, and simplify workflow automation. However, they may impose restrictions on customization, infrastructure control, complex logic, and portability. Therefore, developers should evaluate application requirements, expected workload, integration needs, security, and long-term costs before choosing a low-code platform. For complex requirements, combining Zoho Creator with custom code and external services can provide a practical solution.